

Summary of LHAPDF Set 260000

NNPDF3.0 NLO

General Information

- **Set Description:** NNPDF3.0 NLO global fit, $\alpha_s(M_Z) = 0.118$.
mem=0 $\Rightarrow$ average on replicas; mem=1–100 $\Rightarrow$ PDF replicas.
- **Set Index:** 260000
- **Authors:** NNPDF Collaboration. R.D. Ball, V. Bertone, S. Carrazza, C.S. Deans, L. Del Debbio, S. Forte, A. Guffanti, N.P. Hartland, J.I. Latorre, J. Rojo, M. Ubiali
- **Reference:** arXiv:1410.8849
- **Format:** lhagrid1
- **Data Version:** 2

PDF Characteristics

- **Number of Members:** 101 (1 central + 100 replicas)
- **Particle:** proton (PDG ID = 2212)
- **Flavors included:** $[-5, -4, -3, -2, -1, 1, 2, 3, 4, 5, 21]$
(b, c, s, d, u quarks and antiquarks, plus gluon)
- **QCD Order:** Next-to-Leading Order (NLO)
- **Flavor Scheme:** variable flavor number scheme
- **Number of Active Flavors:** 5
- **Error Type:** replicas (Monte Carlo)

Kinematic Range

- x range: 1.0×10^{-9} to 1.0
- Q range: 1.0 GeV to 10^5 GeV

Mass Parameters (GeV)

- $M_Z = 91.2$
- $m_u = m_d = m_s = 0$ (light quarks neglected)
- $m_c = 1.275$
- $m_b = 4.18$
- $m_t = 173.07$

Strong Coupling α_s

- $\alpha_s(M_Z) = 0.118$
- QCD Order for α_s : NLO
- α_s Type: interpolated (ipol)
- **Tabulated Q values (GeV):**

[1.00, 1.08, 1.17, 1.28, 1.28, 1.39, 1.51, 1.66, 1.83, 2.02,
2.25, 2.51, 2.83, 3.20, 3.65, 4.18, 4.18, 4.77, 5.49, 6.35,
7.39, 8.68, 10.26, 12.22, 14.69, 17.80, 21.77, 26.87, 33.51,
42.22, 53.79, 69.30, 90.37, 119.33, 159.65, 216.54, 297.96,
416.23, 590.67, 852.19, 1250.95, 1869.85, 2848.49, 4426.37,
7022.87, 11387.85, 18891.87, 32098.33, 55918.20, 100000.00]

- **Corresponding α_s values:**

[0.4360, 0.4120, 0.3893, 0.3681, 0.3681, 0.3518, 0.3362, 0.3213, 0.3070, 0.2933,
0.2802, 0.2676, 0.2557, 0.2442, 0.2332, 0.2228, 0.2228, 0.2145, 0.2064, 0.1986,
0.1910, 0.1837, 0.1770, 0.1698, 0.1632, 0.1570, 0.1507, 0.1448, 0.1391, 0.1335,
0.1282, 0.1231, 0.1182, 0.1134, 0.1088, 0.1044, 0.1002, 0.0961, 0.0922, 0.0884,
0.0847, 0.0813, 0.0780, 0.0750, 0.0716, 0.0686, 0.0657, 0.0629, 0.0603, 0.0578]

- $\Lambda_4 = 0.3422, \quad \Lambda_5 = 0.239$

Usage Notes

- Member 0 corresponds to the average (central) PDF.
- Members 1–100 are Monte Carlo replicas for uncertainty estimation.
- Suitable for collider physics applications over a wide kinematic range.
- α_s values are internally interpolated by LHAPDF.